WORKSHEET - NAPLAN Year 9

Number and Algebra
Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 8.75 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. Number and Algebra, NAP-85049

What number makes this number sentence correct?

$1.3 \times \boxed{} = 5.46$

4.2 4.22 6.76 7.098

2. Number and Algebra, NAP-14485

Percy bought 8 packets of cough lollies for $\$18.00$.

The average cost of one packet is

$\$0.45$ $\$2.25$ $\$2.50$ $\$10$

3. Number and Algebra, NAP-79132

Which of the following is equal to 32?

$2^3 \times 2^2$ $2^3 + 2^2$ $3^2 + 2^2$ $3^2 \times 2^2$

4. Number and Algebra, NAP-96743

One billion is one thousand million.

Which of the following is 20 billion?

2×10^{13} 2×10^{10} 2×10^{7} 2×10^{4}

5. Number and Algebra, NAP-02655

$y = 4x^2 - 7$

What is the value of y when $x = 3.1$?

5.4 17.8 31.44 146.76

6. Number and Algebra, NAP-32127

A rule for y in terms of x is $y = 4 - 6x$.

When $x = 2.75$ the value of y is

-16.5 -12.5 -4.5 16.5

7. Number and Algebra, NAP-20295

Which expression is equivalent to $4x^2 - 12x + x^3$?

$4(x^2 - 3x + x^3)$ $4x(x - 3 + x^2)$ $x^2(4 - 12 + x)$ $x(4x - 12 + x^2)$

Worked Solutions

1. Number and Algebra, NAP-85049

$$\begin{aligned} 1.3 \times \boxed{} &= 5.46 \\ \boxed{} &= \frac{5.46}{1.3} \\ &= 4.2 \end{aligned}$$

2. Number and Algebra, NAP-14485

$$\begin{aligned} \text{Price of 1 packet} &= \frac{\$18.00}{8} \\ &= \$2.25 \end{aligned}$$

3. Number and Algebra, NAP-79132

$$\begin{aligned} 2^3 \times 2^2 &= 8 \times 4 \\ &= 32 \end{aligned}$$

4. Number and Algebra, NAP-96743

$$\begin{aligned} \text{20 billion} &= 20 \times 10^3 \times 10^6 \\ &= 2 \times 10^{10} \end{aligned}$$

5. Number and Algebra, NAP-02655

$$\begin{aligned} y &= 4(3.1)^2 - 7 \\ &= 38.44 - 7 \\ &= 31.44 \end{aligned}$$

Worked Solutions

6. Number and Algebra, NAP-32127

$$\begin{aligned} y &= 4 - 6 \times 2.75 \\ &= 4 - 16.5 \\ &= -12.5 \end{aligned}$$

7. Number and Algebra, NAP-20295

$$\begin{aligned} & x\left(4x - 12 + x^2\right) \\ &= 4x^2 - 12x + x^3 \end{aligned}$$